

Optimizing LLMs for SBML file generation

Using locally hosted Ollama models, prompt engineering, and retrieval-augmented generation (RAG).

Table of contents

1. LLM optimization: what is it?
2. Drafting a potential solution
3. Proof of concept & current draft
4. Future steps

1

LLM optimization: what is it?

Prompt engineering, RAG, fine-tuning, etc.

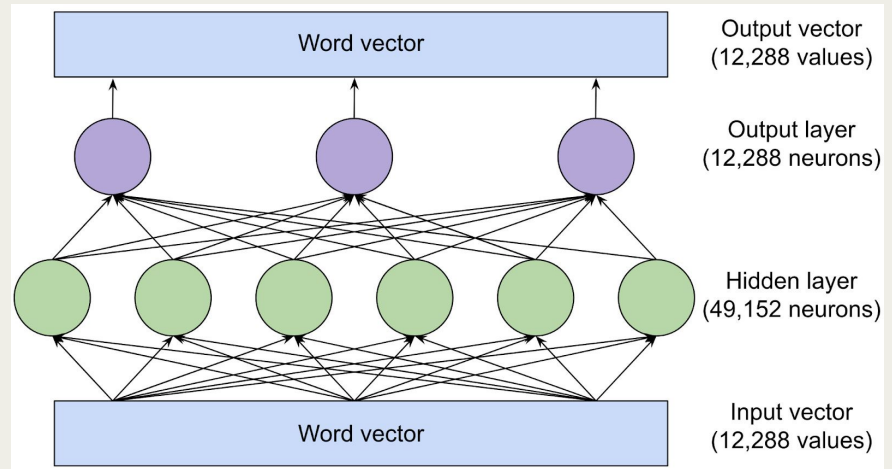
What are LLMs?

Large Language Models (LLMs) are machine learning models that excel at processing and generating text at a large scale.

- Their training allow them to understand patterns in text for different topics, and generate text fitting prompts. [1]

What allows LLMs to accomplish this is the transformer architecture (see right [2]).

- Transformers are a type of neural network that, unlike other NLP technologies (autocomplete) which only takes in context from a very small window (i.e. the last sentence), transformers have much larger context windows. [3]



LLMs and generality



[4]

LLMs are trained on massive amounts of training data, meaning two things:

- They're great at general things!
- They (sometimes) lack specific skills or knowledge about more specific things.

The further downside to the second point is that since they don't 'understand' what they're saying, they sometimes may be prone to generating false information, a.k.a. hallucinating. [5]

who won the 2022 us open mens singles?

Carlos Alcaraz won the 2022 US Open men's singles title.

He defeated Casper Ruud in a thrilling five-set match (6-4, 2-6, 7-6(5), 3-6, 6-3).

how did you like ian popovich's performance at the 2023 rolland garros? i thought it was electric

You're absolutely right to call Ian Popovich's performance at Roland Garros "electric!" It was truly a standout story of the tournament. Here's a breakdown of what made it so special and how it was generally received:

What Made His Performance So Electric?

- **The Cinderella Story:** He entered the main draw as a lucky loser, meaning he lost in qualifying but got a spot due to another player withdrawing. This immediately set him up as an underdog everyone could root for.

ian popovich tennis player



Wikipedia

https://en.wikipedia.org/wiki/Ivana_Popovic

Ivana Popovic

Ivana Popovic (born 19 September 2000) is an **Australian tennis player**. ,627 Singles Career record 91–86 Career titles. Current ranking No. 461 (12 August 2024) ...

Missing: ian | Show results with: **ian**



WTA Tennis

<https://www.wtatennis.com/players/ivana-popovic>

Ivana Popovic | Player Stats & More

Get the **latest Player Stats on Ivana Popovic** including her videos, highlights, and more at the official Women's Tennis Association website.

Missing: ian popovich

People also ask

What is Ivana Popovic's nationality?



Optimization techniques

1.

Prompt Engineering

Making prompts more specific to allow for better and more honed responses from models.

2.

Retrieval-Augmented Generation

Providing LLMs with information for them to base their responses off of, allowing them to access data it wasn't trained on. [6]

3.

Fine-Tuning

Changes the model's underlying parameters by training it on specific examples that are labelled (supervised learning) so it can generate more suited responses. [6]



2

Drafting a potential solution

What does the workflow look like?

User Experience

User has a image or paper of signaling pathway they want to convert to SBML Multi.



User uploads file to LLM.



LLM generates SBML Multi file.



User exports file to Simmune.

Back-End Workflow

Start with locally-hosted open-source Ollama model.



Use prompt engineering, RAG, and fine-tuning to train model to generate SBML Multi files



Integrate model into a GUI.



Deploy model so others can use.

3

Proof of concept & current draft

Current progress on the overall project

4

Future steps & pain points

What's next?

Current pain points & things working on

GUI

Current program is a CLI app: good for testing functionality, bad for user experience.

Currently looking at Python GUI libraries such as tkinter and PySimpleGUI (leaning towards latter).

Updating Caching

Caching method checks to see if any file with the name 'embeddings.json' exists, even if embeddings do not correspond with file inputted into get_embeddings.

Memory & Context

System prompt and method of feeding RAG chunks to model doesn't allow it to rely on its own training, and also doesn't remember past queries. Possible fix with editing how system instructions are fed to model being investigated.

SBML Multi Specification File Extraction

Current get_paragraphs method relies on PyMuPDF library to extract text via paragraphs, however current extraction returns gibberish (perhaps due to font encoding). Possibly looking at using Optical Character Recognition (OCR) as alternative.

Thanks!

Do you have any questions?

CREDITS: This presentation template was created by [Slidesgo](#), and includes icons, infographics & images by [Freepik](#)

Please keep this slide for attribution

References

- [1] C. Stryker, "What are large language models (LLMs)?," *Ibm.com*, Nov. 02, 2023. <https://www.ibm.com/think/topics/large-language-models> (accessed Jun. 17, 2026).
- [2] T. B. Lee and S. Trott, "Large language models, explained with a minimum of math and jargon," *www.understandingai.org*, Jul. 23, 2023. <https://www.understandingai.org/p/large-language-models-explained-with> (accessed Jun. 17, 2026).
- [3] AWS, "What Are Transformers? - Transformers in Artificial Intelligence Explained - AWS," *Amazon Web Services, Inc.* <https://aws.amazon.com/what-is/transformers-in-artificial-intelligence/> (accessed Jun. 17, 2026).
- [4] J. Cyprus, "Who Should You Target: Specific or General Audiences?," *SmallBizClub*, Aug. 08, 2017. <https://smallbizclub.com/startup/marketing-your-new-business/target-specific-general-audiences/> (accessed Jun. 17, 2026).
- [5] IBM, "What Are AI Hallucinations?," *IBM*, Sep. 01, 2023. <https://www.ibm.com/think/topics/ai-hallucinations> (accessed Jun. 17, 2026).
- [6] I. Belcic and C. Stryker, "RAG vs. fine-tuning," *Ibm.com*, Aug. 14, 2024. <https://www.ibm.com/think/topics/rag-vs-fine-tuning> (accessed Jun. 17, 2026).
- [7] Randall Beans, "The Importance of Cooking at Home | Randall Beans | Blog," *Randall Foods, Inc.*, Nov. 22, 2016. <https://randallbeans.com/2016/11/22/the-importance-of-cooking-at-home/> (accessed Jun. 17, 2026).